

Antonio Carmagnani

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PROFILE

Experience in computational design and data analysis for real estate and cities, spanning academic research and architecture practice. Recently completed an M.Arch. in Architecture Computation at The Royal Danish Academy, with a focus on computational and data methods for real estate and urban development. Hands-on skills span Python, QGIS, and machine learning, parametric workflows in Grasshopper, and building data tools that turn spatial and financial data into decisions. Motivated by work where the built environment, technology, and data intersect.

EXPERIENCE

Project Assistant in International Infrastructure Projects @ Sweco Danmark A/S

Copenhagen, DK

02.2026 - 07.2026

- Support consulting teams delivering internationally funded infrastructure projects in water, energy, and transport across emerging and developing economies, financed by development banks including the World Bank, EBRD, IFDK, KfW and EIB.
- Authored climate risk assessments for major water supply projects following JASPERS and EBRD climate-proofing guidance, screening hazards against projections data and rating risk by exposure and vulnerability for each asset.
- Built geospatial databases for water and infrastructure projects in QGIS, compiling georeferenced asset locations from agency data, satellite imagery, and open sources for socio-spatial analysis.

Intern in Responsible Business Conduct @ Ramboll Management Consulting

Copenhagen, DK

05.2025 - 08.2025

- Conducted human rights due diligence for corporate clients, applying the UN Guiding Principles on Business and Human Rights and OECD Guidelines to assess salient risks across global supply chains.
- Mapped the global supply chains of 20+ critical raw materials under the EU Critical Raw Materials Act, profiling producing companies, sectors, and geographies from authoritative industry databases.
- Developed a client-facing overview of the firm's real estate ESG due diligence services, centered on building energy and decarbonization assessments aligned with EU building performance regulations.
- Built a fully automated risk-assessment system in Excel, consolidating risk data across supply chains into a single dashboard that updated salient risks dynamically.
- Integrated AI research tools into due diligence workflows to source and cite human rights risk data at scale across high research volumes.

Architect @ Isay Weinfeld Architects

São Paulo, BR

02.2023 - 08.2024

- Delivered high-end residential, hotel, and mixed-use projects across South America, North America, Europe, the Middle East, and Asia at one of Brazil's most prominent architecture studios.
- Built a Grasshopper plugin in Python that generates bespoke staircases from a floor plan, cutting staircase modeling from days to around 20 minutes and enabling far more design iterations and geometric complexity.
- Presented the tool in a studio workshop and supported its adoption by colleagues across 10+ projects.
- Scripted parametric and computational models in Rhino and Grasshopper for geometrically complex forms, supporting design development across projects.
- Ran daylight and solar-exposure simulations with Ladybug in Grasshopper to drive massing and orientation decisions, quantifying light incidence across building layouts.

Trainee in Real Estate Product Development @ OSPA

São Paulo, BR

02.2022 - 10.2022

- Produced 15+ feasibility studies for mixed-use and residential developments across Brazil and the US (São Paulo, Rio de Janeiro, Porto Alegre, Miami) for developer clients.
- Developed volumetric massing studies in Revit, resolving circulation, unit layouts, ground-floor active frontage, and parking against zoning metrics.
- Ran preliminary environmental and solar studies to inform massing and layout decisions.

Co-Founder @ VIZU

São Paulo, BR

02.2022 - 10.2022

- Co-founded and ran an architectural visualization studio, managing 10+ clients, project scoping and delivery, and hiring staff.
- Produced architectural renders, humanized plans, and visualizations to communicate residential and commercial designs to investors and end buyers.

Architect @ Felipe Hess Architects

São Paulo, BR

07.2021 - 02.2022

- Developed high-end residential projects in São Paulo and Rio de Janeiro from preliminary studies through technical documentation, working in Revit, AutoCAD, and SketchUp.
- Ran environmental and daylight studies to inform early design decisions.

Intern in Project Management @ Itaú-Unibanco**São Paulo, BR**

08.2020 - 07.2021

- Reviewed technical documentation from architecture and engineering firms prior to construction authorization, checking drawings against building, accessibility, and fire-safety norms (NBR standards).
- Assessed the environmental and energy performance of buildings the bank was acquiring or leasing, reviewing consumption data, efficiency ratings, and certifications as part of internal due diligence.
- Built a VBA-powered Excel dashboard tracking the bank's corporate furniture inventory across locations, replacing a manual process and speeding up procurement for new-building fit-outs.

EDUCATION M.Arch. in Architecture Computation @ The Royal Danish Academy**Copenhagen, DK**

07.2024 - 07.2026

Semester Programme in Building Technology @ Architecture Association**London, UK**

07.2022 - 01.2023

Summer School in Parametric Design @ Architecture Association**London, UK**

06.2019 - 07.2019

B.Arch. in Architecture and Urbanism @ Mackenzie University**São Paulo, BR**

07.2018 - 12.2023

PROJECTS Rammed Blue Biomass & Computational Optimization @ The Royal Danish Academy**Copenhagen, DK**

- Developed a new construction technique and material system using blue biomass (seashells and algae), applying LCA metrics and parametric simulations to optimize structural performance and environmental impact.
- Project was selected as a semester highlight and served as the material foundation for the school's installation at the Venice Architecture Biennale 2025, presented in collaboration with CITA, GXN, and 3XN at the Arsenale.

Venice Architecture Biennale 2025 @ The Royal Danish Academy, CITA, GXN, 3XN**Venice, IT**

- Contributed to the Matanzas panel by developing geospatial mappings to analyze local material sources and supply flows, producing cartographic work that informed the site's material research.
- Contributed to the Veneto panel by producing the physical exhibition panels displayed in the 7-metre installation at the Arsenale, which used the blue biomass composite technique developed during the semester project.

Embodied Carbon Toolkit**Copenhagen, DK**

- Collaborated with a Copenhagen-based LCA startup to design and deliver a Revit-based toolkit for embodied carbon calculations, integrating the platform's GWP values for A1-A3 and A4 stages.
- Built a pre-configured Revit schedule that computes embodied carbon with built-in documentation for adoption by architects.

Master's Thesis, Computational Urban Analysis for Intensification Strategy @ The Royal Danish Academy**Copenhagen, DK**

- Developed a computational framework to identify underused spaces in the historic city center of São Paulo, Brazil, combining urban morphology analysis, land use data, and social vulnerability indicators.
- Built an algorithm that uses urban data, including building typology, use patterns, regulatory conditions, and building use, to generate programmatic and morphological typologies for urban intensification, aligned with the city's existing urban policies.
- Integrated multiple geospatial datasets in QGIS to map spatial relationships between buildings, zoning conditions, and social context at the neighborhood scale.

Real Estate Development Recommendation Tool @ The Royal Danish Academy**Copenhagen, DK**

- Built a machine learning tool that generates site-specific development recommendations for buildings in Copenhagen by integrating financial, social, sustainability, and geospatial data.
- Designed a user interface allowing different stakeholders to adjust impact drivers and weighting, with predictive outputs aligned to ESG priorities and local development conditions.

SKILLS

- Data & Analysis: Python, QGIS, SQL, Power BI, Excel / VBA, HTML
- Digital Tools: Rhino + Grasshopper, Revit, AutoCAD, Adobe Suite (InDesign, Illustrator, Photoshop)
- ESG & Sustainability: CSRD, Double Materiality, EU Taxonomy, UNGPs, OECD Guidelines, HRDD, HRIA, LCA (Life Cycle Assessment)
- Real Estate & Urban: Real estate feasibility modeling, Zoning and land use regulatory analysis, Urban morphology and spatial analysis
- Languages: English (fluent), Portuguese (native), Spanish (fluent), Danish (limited working proficiency), Italian (limited working proficiency)